Neural Processing Letters (2026) 58:44
https://doi.org/10.1007/s11063-026-11851-4

Lightweight Multi-channel Fall Detection on Smartphone Accelerometers with Probabilistic Calibration and Event-Level Evaluation

Zhiyuan Yang[1,4,5] · Liang Li[5] · Zhiqiang Ren[4,5] · Qinghua Meng[1] · Yijia Xue[1,4,5] · Can Bu[3,4,5] · Liangze Tao[3,4,5] · Kai Xiong[4,5] · Dongren Liu[1,2]

Received: 2 February 2026 / Accepted: 19 April 2026 / Published online: 13 May 2026
© The Author(s) 2026

Abstract
Accurate and energy-efficient fall detection on commodity smartphones is crucial for long-term monitoring of older adults and high-risk populations. However, most existing methods emphasize window-level accuracy and F1 score, while largely overlooking probabilistic calibration and event-level behavior that are critical for real-world deployment. We propose LFNET, a lightweight multi-channel convolutional network for fall detection using a single waist-mounted smartphone accelerometer. LFNET integrates multiscale depthwise separable convolutions with efficient channel attention, and jointly leverages tri-axial accelerations and their magnitude to capture both posture dynamics and impact intensity. On the UniMiB-SHAR dataset, LFNET attains competitive or superior F1 and AUPRC on both the binary fall detection task (AF2) and the 17-class activity recognition task (AF17), while using only a fraction of the parameters and FLOPs of larger CNN and Transformer baselines. To better match deployment requirements, we apply temperature scaling and sensitivity-constrained threshold selection to calibrate predicted probabilities and choose operating points tailored to fall-risk preferences. Using semi-synthetic continuous streams constructed from UniMiB-SHAR segments, we further design an event-level post-processing pipeline combining K-consecutive positives, hysteresis thresholds, gap merging, and refractory periods. Event-level experiments show that, under a fixed configuration, LFNET improves event sensitivity from 90.1% to 96.4% over a CNN baseline, while reducing false alarms from 1.4 to 0.2 FA/h and shortening median detection delay from 1.4 s to 0.8 s. These results demonstrate that the proposed framework not only performs well on standard window-level metrics, but also offers favorable false-alarm–delay trade-offs for smartphone-based, long-term fall detection in simulated monitoring scenarios.

Keywords Fall detection · Smartphone · Multi-channel convolutional network · Probabilistic calibration · Event-level post-processing

1 Introduction

Falls are one of the leading causes of injury, disability, and mortality among older adults and other vulnerable populations, and they impose a rapidly increasing burden on acute care,

Extended author information available on the last page of the article

long-term care, and informal caregiving systems worldwide [1–3]. With population ageing and the rising proportion of older adults living alone, there is an urgent need for fall detection solutions that can be deployed at scale in everyday environments, while remaining low-cost, energy-efficient, and minimally intrusive [4, 5]. Against this backdrop, timely and reliable fall detection in home and community settings has become a central topic in public health, gerontechnology, and digital health research.

Conventional fall monitoring approaches rely largely on human supervision, fixed alarm devices, or video surveillance. Although such solutions can be effective in controlled environments such as hospitals or residential care facilities, they suffer from important limitations in home and community settings: installation and maintenance are costly, spatial coverage is restricted, performance is highly dependent on user adherence, and continuous video monitoring raises substantial privacy and acceptability concerns [6]. In contrast, wearable and mobile sensing technologies offer a promising alternative for continuous monitoring of human motion. In particular, inertial measurement units (IMUs) comprising accelerometers and gyroscopes can capture human movement with high temporal resolution, providing rich time-series signals for human activity recognition (HAR) and fall detection [7, 8].

Among the various wearable and mobile platforms, smartphones offer several unique advantages as fall detection devices. First, smartphones are already pervasive and strongly integrated into users' daily routines, eliminating the need to introduce additional dedicated devices and thereby reducing deployment barriers and cost [9, 10]. Second, modern smartphones typically embed tri-axial accelerometers and gyroscopes with sampling rates of at least 50 Hz, which are sufficient to capture key motion patterns such as gait changes, postural transitions, and impact events [11, 15]. Third, contemporary smartphones provide non-trivial on-device computational resources and a mature software ecosystem, enabling real-time inference and local alerting at the edge, while also supporting data upload, remote updates, and multi-level alarm strategies via cellular and local networks [13, 14]. Finally, mobile operating systems now offer relatively mature permission and privacy management mechanisms, which facilitate privacy-preserving data collection and local processing under appropriate regulatory frameworks. These characteristics make smartphone-based fall detection systems an attractive option for real-world deployment and position smartphone IMU datasets as ideal testbeds for methodological research [15].

A large body of work has investigated sensor-based fall detection and HAR. From a methodological perspective, early studies predominantly adopted traditional machine learning pipelines: hand-crafted time- and frequency-domain features (e.g., mean, variance, energy, spectral entropy) were extracted from fixed-length sliding windows, followed by classification using support vector machines (SVM), k-nearest neighbours (kNN), or random forests [12, 16, 17]. These approaches are relatively simple and computationally efficient, but they depend heavily on domain expertise for feature design, often struggle to capture complex movement patterns, and are sensitive to sensor placement and inter-subject variability. With the success of deep learning in sequence modelling, recent work has increasingly shifted towards end-to-end models such as convolutional neural networks (CNNs), temporal convolutional networks (TCNs), long short-term memory (LSTM) networks, and Transformer-based architectures, which have achieved substantially better window-level recognition performance on public HAR and fall detection datasets [18, 19].

Despite this progress, several important challenges remain. First, many existing models are highly sensitive to sensor orientation, wearing position, and cross-domain distribution shifts; in practice, variations in smartphone orientation, carrying location, and individual gait patterns can lead to pronounced degradation in performance [20]. Second, numerous state-of-the-art architectures are structurally complex and parameter-heavy, limiting their suitability

for resource-constrained mobile devices and battery-powered continuous monitoring scenarios [18, 19]. Third, most current studies focus on window-level classification metrics, whereas clinically and operationally relevant outcomes such as event-level sensitivity, false alarm rate, and detection latency over simulated long-term monitoring remain comparatively understudied [18–20]. Addressing these gaps is crucial for translating smartphone-based fall detection algorithms from benchmark datasets to robust, deployable systems in real-world settings.

Building on the above motivation, this study focuses on the UniMiB-SHAR dataset of smartphone-based IMU signals and proposes a lightweight multi-channel deep network together with a complete evaluation pipeline that bridges window-level scores and event-level alarms [21]. The main contributions are as follows:

1. Multi-channel input design: We utilize a multi-channel representation that jointly incorporates the three Cartesian axes and the acceleration magnitude to enhance robustness against sensor orientation variations.
2. Lightweight multi-channel CNN architecture (LFNET): We propose a multi-channel lightweight convolutional neural network, termed LFNET, combining depthwise separable convolutions and efficient channel attention (ECA) to achieve high feature discrimination under strict computational budgets.
3. Calibrated training and risk-aware thresholding: We apply temperature scaling and select decision thresholds subject to a high-sensitivity constraint (≥ 0.95), reflecting the high clinical cost of missed falls.
4. Event-level evaluation pipeline: We transition from standard window-level metrics to event-level evaluations on semi-synthetic continuous streams, assessing real-world viability via false-alarm and delay trade-offs.

The remainder of this paper is organized as follows. Section 2 describes the dataset, preprocessing pipeline, and model architecture in detail. Section 3 reports the main experimental results, ablation studies, and both statistical and engineering-oriented metrics. Section 4 discusses key findings, compares our approach with related work, and outlines limitations. Section 5 concludes the paper and provides directions for future research.

2 Materials and Methods

2.1 Dataset and Data Partitioning

This study is conducted on the UniMiB-SHAR dataset. UniMiB-SHAR was collected at the University of Milano-Bicocca using a single Android smartphone (Samsung Galaxy Nexus I9250) equipped with a tri-axial accelerometer, placed in the front trouser pocket of the participants (with left and right pockets used in approximately equal proportions across trials). The device recorded acceleration signals at a sampling rate of approximately 50 Hz.

The dataset comprises recordings from 30 healthy volunteers (6 males and 24 females), aged between 18 and 60 years, and contains a total of 11,771 activity segments. Each segment corresponds to a fixed-length time window of about 3 s centred on a single activity or fall event, and is represented as a sequence of accelerometer triplets that encode the linear acceleration components of the smartphone along the three Cartesian axes. An illustrative example of the signal segmentation and sensor axes is shown in Fig. 1.

To balance fine-grained recognition with practical deployment requirements, we define two main tasks:

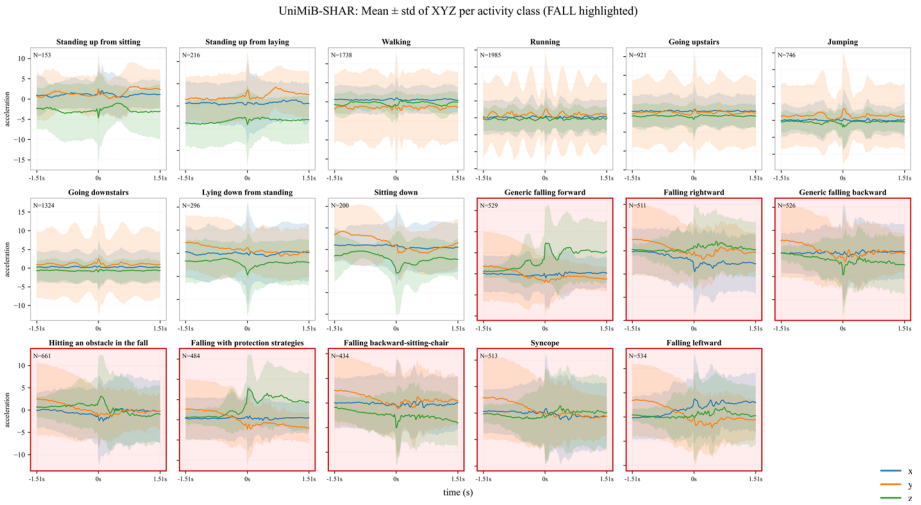


Fig. 1 Class-wise mean and standard deviation of the tri-axial acceleration signals in the UniMiB-SHAR dataset. Here, N denotes the number of samples in each class, and classes highlighted in red correspond to fall events

- A multi-class task involving 17 fine-grained activity and fall classes.
- A binary task distinguishing between fall and non-fall events.

UniMiB-SHAR provides a subject identifier for every segment, enabling subject-wise data partitions that prevent samples from the same person appearing in both training and test sets and thus yield a more realistic assessment of cross-subject generalization. In this study we preserved each segment's subject ID during preprocessing and used it to implement two evaluation protocols.

The first protocol is a fixed subject-wise 70:15:15 split: all subjects are randomly assigned to train, validation, and test sets based on their unique user identifiers, ensuring no data leakage across subsets.

The second protocol is leave-one-subject-out (LOSO): in each fold, all segments from one subject form the test set while segments from the remaining subjects are used for training and validation; the procedure is repeated for every subject. LOSO simulates the strictest "completely new user" generalization scenario.

Model training, hyperparameter tuning and probability calibration are performed using the training and validation partitions; test-set subjects are used only for final evaluation.

2.2 Preprocessing and Multi-Channel Input Representation

In the UniMiB-SHAR dataset, each sample corresponds to a fixed-length segment of an accelerometer time series, which can be formalized as

$$\mathbf{a}(t) = \big(a_x(t), a_y(t), a_z(t)\big), \quad t = 1, \ldots, T. \tag{1}$$

where $a_x(t)$, $a_y(t)$, and $a_z(t)$ denote the linear acceleration components of the smartphone along the three Cartesian axes, and T is the number of time steps per sample. Building on the raw tri-axial acceleration, we construct a multi-channel input that incorporates the

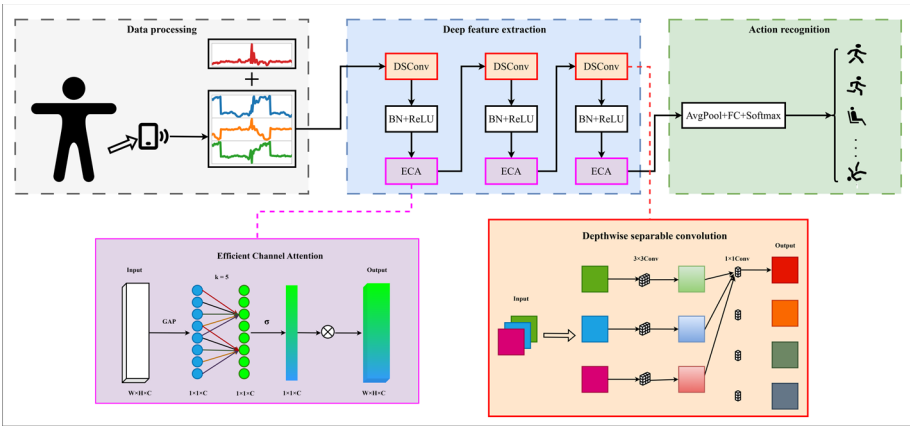


Fig. 2 LFNET model structure diagram

signal magnitude and apply per-channel normalization to improve robustness to variations in wearing posture and inter-subject differences.

First, to explicitly capture the overall acceleration intensity, we derive a magnitude channel in addition to the three axes:

$$a_m(t) = \sqrt{a_x^2(t) + a_y^2(t) + a_z^2(t)}, \quad t = 1, \ldots, T. \tag{2}$$

Compared with using only the three Cartesian components, introducing the magnitude channel helps attenuate the effect of changes in smartphone orientation. While extracting magnitude is an established and widely adopted strategy in human activity recognition to improve rotational invariance, our network explicitly integrates it as an augmented input channel. When the device rotates within the pocket, the individual axis components may change substantially, whereas the magnitude is invariant to pure rotations and thus provides a more stable representation of global motion intensity and impact characteristics. Consequently, each sample is finally represented as a multi-channel time series of length T:

$$\mathbf{X} = \left[a_x, a_y, a_z, a_m\right] \in \mathbb{R}^{T \times 4} \tag{3}$$

To mitigate the impact of sensor bias, differences in movement amplitude across subjects, and variations in recording conditions, we perform per-channel z-score normalization on the input. This transformation rescales each channel to have zero mean and unit variance, thereby reducing scale discrepancies induced by inter-subject variability and allowing the model to focus on relative temporal patterns that are more likely to generalize across users and deployment scenarios.

2.3 Lightweight Deep Network Architecture

Building on the above preprocessing and multi-channel input representation, we design a multi-channel lightweight convolutional neural network to balance fall-detection performance against model complexity and computational cost (see Fig. 2). The architecture is constructed on a unified multi-channel input and follows the principle of "sufficient temporal pattern modelling, controlled parameter budget, and deployment friendliness."

2.3.1 Depthwise Separable Convolutions

To reconcile recognition performance with the computational constraints of mobile deployment, we extensively adopt depthwise separable convolutions in the convolutional backbone in place of standard convolutions. For a conventional convolutional layer with C_{in} input channels, C_{out} output channels, and a kernel size of $k_t \times k_s$ (where k_s denotes the kernel extent along the stacked multi-channel layout axis), the number of parameters is:

$$\text{Params}_{conv} = C_{in} \times C_{out} \times k_t \times k_s \tag{4}$$

In contrast, a depthwise separable convolution decomposes the operation into two steps: a channel-wise spatial convolution (depthwise convolution) followed by a 1×1 pointwise convolution. Its parameter count is

$$\text{Params}_{dw+pw} = C_{in} \cdot k_t \cdot k_s + C_{in} \cdot C_{out} \tag{5}$$

The corresponding compression ratio with respect to a standard convolution is approximately

$$\frac{\text{Params}_{dw+pw}}{\text{Params}_{conv}} \approx \frac{1}{C_{out}} + \frac{1}{k_t k_s} \tag{6}$$

Under typical settings in our network, this reduces the parameters and floating-point operations of a single convolutional layer to about one-fifth of the original. As a result, we can stack more layers or use wider channels under the same computational budget, thereby enhancing the representational capacity of the model without violating resource constraints.

2.3.2 ECA Channel Attention

Although depthwise separable convolutions substantially reduce computational cost, effectively exploiting the complementary information encoded in different channels remains crucial for achieving strong recognition performance under a limited channel budget. To this end, we incorporate Efficient Channel Attention (ECA) modules at the output of each convolutional stage to perform lightweight channel-wise re-calibration.

The core idea of ECA is to model inter-channel dependencies using a local one-dimensional convolution, without introducing fully connected layers or performing channel dimensionality reduction. Formally, given a feature map

$$F \in \mathbb{R}^{C \times T' \times M'} \tag{7}$$

ECA first applies global average pooling over the T' and M' dimensions of F to obtain a channel descriptor $S \in \mathbb{R}^C$. A one-dimensional convolution of kernel size k is then applied along the channel dimension of S, yielding an attention vector $\alpha \in \mathbb{R}^C$. After a sigmoid activation to map the coefficients to the interval $(0, 1)$, the original feature map is re-weighted channel-wise:

$$F'_c = \alpha_c \cdot F_c, \quad c = 1, \ldots, C \tag{8}$$

Here, the ECA module introduces only a negligible number of additional parameters (with kernel size k typically set to 3 or 5) and minimal computational overhead, yet it effectively enhances channels that are informative for the target task while suppressing redundant or noisy ones. In our network, ECA modules are inserted after the convolutional blocks of each stage so that, as the network depth increases, channel attention can progressively focus on dynamic patterns that are strongly associated with falls and related activities across multiple temporal scales.

2.3.3 Output Layer

After several convolutional and attention-recalibration stages, the final feature map is fed into a global average pooling (GAP) layer, which aggregates the T' and M' dimensions into a fixed-length feature vector:

$$z = \mathrm{GAP}(F_c') \in \mathbb{R}^D \tag{9}$$

This vector is then passed through a single fully connected layer to obtain the class logits:

$$o = Wz + b \in \mathbb{R}^K \tag{10}$$

$$\hat{p}_k = \frac{\exp(o_k)}{\sum_{j=1}^{K} \exp(o_j)}, \quad k = 1, \ldots, K \tag{11}$$

where K is the number of classes ($K = 17$ for the AF17 task and $K = 2$ for the AF2 task), and $\hat{p}$ denotes the predicted class-probability distribution.

For the binary AF2 task, the dataset exhibits a noticeable class imbalance: among the 11,771 total segments, there are 4192 fall events (minority) and 7579 non-fall ADL events (majority). Given this imbalance in AF2 and the frequency disparities across the 17 categories in AF17, we adopt Focal Loss instead of the standard cross-entropy to reduce the dominance of easily classified majority-class samples and to emphasize hard and minority-class examples.

For the binary task, let $y = 1$ denote the fall (positive) class and $y = 0$ denote the non-fall (negative) class, and let $p = P(y = 1 \mid x)$ be the model's predicted probability for the positive class. The per-sample Focal Loss is defined as:

$$\mathrm{FL}(y, \hat{p}) = -\alpha y (1 - \hat{p})^{\gamma} \log(\hat{p}) - (1 - \alpha)(1 - y)\hat{p}^{\gamma} \log(1 - \hat{p}) \tag{12}$$

where $\alpha \in (0, 1)$ is a class-balancing factor (e.g., to upweight the positive class), and $\gamma \geq 0$ is the focusing parameter. When $\gamma = 0$ and $\alpha = 0.5$, Focal Loss degenerates to the standard binary cross-entropy.

For the multi-class task, let $y \in \{0, 1\}^K$ be the one-hot ground-truth label vector and $\hat{p}$ the predicted probability vector. The multi-class Focal Loss is given by:

$$\mathrm{FL}(y, \hat{p}) = -\sum_{k=1}^{K} \alpha_k y_k (1 - \hat{p}_k)^{\gamma} \log(\hat{p}_k) \tag{13}$$

where α_k controls the relative weight of class k.

2.3.4 LFNET Configuration Details

To ensure full reproducibility, the detailed layer-by-layer configuration of LFNET is summarized in Table 1. The network consists of three cascaded feature-extraction stages. As illustrated in Fig. 2, each stage strictly comprises a depthwise separable convolution (DSConv) block (using a 3×3 depthwise kernel and a 1×1 pointwise kernel), a batch normalization (BN) and ReLU layer, and an ECA module with kernel size $k = 5$. The extracted features are ultimately mapped to classification logits via global average pooling (GAP) and a fully connected (FC) layer. In our implementation, each segment has $T = 150$ time steps, corresponding to a 3-second window at 50 Hz. We set the stage-wise channel widths to $C_1 = 64$, $C_2 = 128$, and $C_3 = 256$. Each depthwise convolution uses stride $s = 1$ on the joint time–channel grid with padding chosen so that the temporal length remains equal to the input segment length T at the output of every stage.

Table 1 Layer-by-layer architectural specification of LFNET. T denotes the number of time steps in each sliding-window segment ($T = 150$ for a 3-second window at 50 Hz). DW and PW denote depthwise and pointwise convolutions, respectively. Implemented channel widths are $C_1 = 64$, $C_2 = 128$, and $C_3 = 256$

Stage	Layer	Kernel / stride	Output channels	Feature map
Input	Multi-channel signal	—	4	$T \times 4$
Stage 1	DSConv	DW: 3×3, $s = 1$; PW: 1×1	C_1	$T \times C_1$
	BN + ReLU	—	C_1	$T \times C_1$
	ECA	1D conv, $k = 5$	C_1	$T \times C_1$
Stage 2	DSConv	DW: 3×3, $s = 1$; PW: 1×1	C_2	$T \times C_2$
	BN + ReLU	—	C_2	$T \times C_2$
	ECA	1D conv, $k = 5$	C_2	$T \times C_2$
Stage 3	DSConv	DW: 3×3, $s = 1$; PW: 1×1	C_3	$T \times C_3$
	BN + ReLU	—	C_3	$T \times C_3$
	ECA	1D conv, $k = 5$	C_3	$T \times C_3$
Output	GAP	—	C_3	$1 \times C_3$
	FC	—	K (2 or 17)	$1 \times K$

2.4 Probability Calibration and Threshold Selection

The softmax outputs of deep networks are often miscalibrated, exhibiting either over-confident or under-confident behaviour; for example, samples predicted with a probability of 0.9 may have a true empirical accuracy substantially lower than 90%. To mitigate this issue, we apply temperature scaling to post-hoc calibrate the output probabilities of the AF2 head.

Let the uncalibrated AF2 logits be denoted as $z = [z_0, z_1]$, where z_0 corresponds to the non-fall class and z_1 to the fall class. Temperature scaling divides the logits by a positive scalar temperature $T > 0$ before the softmax, yielding the calibrated probability distribution:

$$\hat{p}_T = \mathrm{softmax}(z/T)$$

The temperature T is learned on the AF2 validation set by minimizing the negative log-likelihood (NLL). During this optimization, only T is updated; the backbone network and the classification head remain fixed. As a result, temperature scaling does not change the predicted class labels (i.e., the argmax over logits), but only adjusts the confidence values. The fitted temperature T is then held fixed and reused for both the AF2 test set and the continuous-stream evaluations, without any recalibration on test data.

For threshold selection in fall detection, we adopt a high-sensitivity constrained strategy. After temperature scaling, let $s^{(i)} = \hat{p}_T(y = 1 \mid x^{(i)})$ denote the calibrated fall probability for sample i on the AF2 head. To reflect the "high cost of missed falls" design principle, we prioritize sensitivity when choosing the decision threshold on the validation set. For any threshold $\tau \in [0, 1]$, we can compute its validation sensitivity $\mathrm{Sensitivity}(\tau)$. Given a target sensitivity lower bound τ_s, we scan a range of candidate thresholds on the validation set, collect all τ that satisfy $\mathrm{Sensitivity}(\tau) \geq \tau_s$ and select the smallest one, denoted $\tau^* = \min \tau$, as the final decision threshold. This procedure guarantees that the model achieves at least the desired high sensitivity on the validation set, while using the lowest possible threshold to further reduce the risk of missed falls. The additional false alarms induced by a lower τ are subsequently controlled at the event level via post-processing strategies such as K-consecutive rules, hysteresis, and refractory periods.

2.5 Event-Level Post-Processing and Semi-Synthetic Stream Evaluation

2.5.1 Construction of Semi-Synthetic Continuous Streams

In UniMiB-SHAR, each 3s segment is collected independently without temporal continuity, so segment-level evaluation alone cannot characterize event-level performance during long-term monitoring. To approximate continuous use, we construct semi-synthetic continuous streams.

For each subject, we randomly sample and concatenate activity of daily living (ADL) segments from that subject's pool to form an ADL-only stream of approximately 1 h, treated as a "daily mixed activity" background. To reduce abrupt jumps at segment boundaries, we apply a short linear cross-fade over a few samples between neighbouring segments, yielding smoother, more realistic accelerometer traces.

On each ADL stream, we then insert 10– 20 fall segments at uniformly random, non-overlapping time positions, obtaining annotated semi-synthetic streams. This density provides sufficient events for stable estimation of event-level sensitivity and delay within a feasible simulation length, while avoiding unrealistically frequent falls. Representative examples of a 1-hour semi-synthetic stream and a 1-minute zoomed-in segment are shown in Fig. 3.

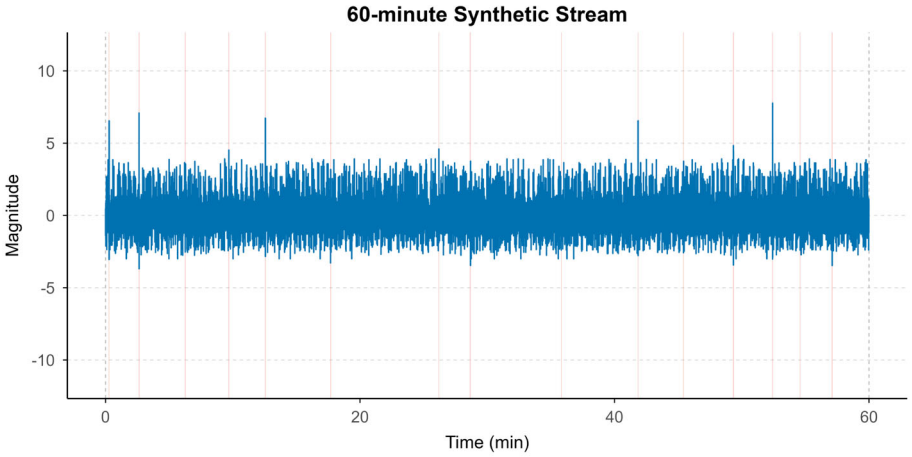
2.5.2 Online Sliding-Window Inference

On each semi-synthetic stream, we apply a sliding window of length $L = 3$ s, with a strictly fixed step size of $\Delta = 0.5$ s. For every window, the AF2 head produces raw logits, which are calibrated using the previously estimated temperature T^* to obtain the fall probability. This probability is compared with the validation-derived decision threshold τ^*, yielding a binary window-level decision (fall / non-fall). The resulting temporal sequence of binary predictions serves as the input to event-level post-processing.

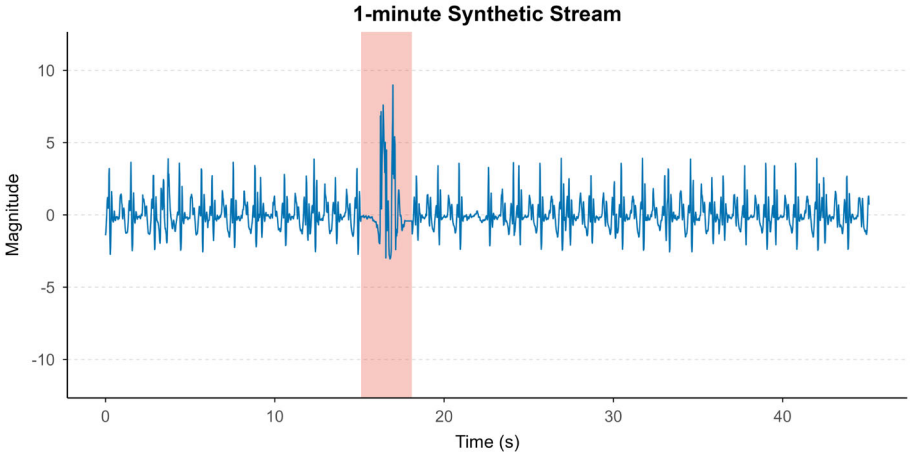
2.5.3 Event-Level Post-Processing

To obtain stable event-level alarms, we apply four simple rules to the window-level binary sequence:

- K-consecutive positives: A fall is triggered only if at least K consecutive windows are positive. Given our 3 s window and the 0.5 s step size, the overlap between consecutive windows is exceptionally high (83.3%). This intentional design ensures that the sudden, sharp impact signal of a fall is captured across multiple sliding windows, making the K-consecutive rule ($K = 5$) effective at suppressing isolated noise while reliably confirming true events.
- Hysteresis thresholds: events are opened when the probability exceeds τ_{on} and closed only when it remains below τ_{off}, reducing chattering near the decision boundary.
- Gap merging: positive clusters separated by a time gap shorter than a preset threshold are merged into a single fall episode.
- Refractory period: after a detected fall, a refractory interval is enforced during which new positives are ignored, limiting repeated alarms due to post-fall movements or caregiving actions.



(a) 1-hour semi-synthetic stream



(b) 1-minute zoomed-in segment

Fig. 3 Representative examples of streams

Default parameter values are tuned on validation streams and, in our setting, provide a reasonable trade-off among event-level sensitivity (Event Sens), false alarms per hour (FA/h), and detection delay (Delay).

2.6 Training Strategy and Evaluation Metrics

All experiments use a unified training setup with the AdamW optimizer, a constant learning rate of 5×10^{-4}, a weight decay of 1×10^{-4}, and a batch size of 128 for up to 100 epochs. We fix random seeds to ensure reproducibility. To address class imbalance, we configured the Focal Loss with the focusing parameter $\gamma = 2.0$ for all tasks. For the binary AF2 task, the balancing factor α was strictly set to 0.75 for the fall class and 0.25 for the non-fall class. For the multi-class AF17 task, class-specific weights α_k were computed dynamically using

inverse class frequencies as $\alpha_k = N_{\text{total}}/(17 \times N_k)$, where N_k denotes the number of training segments with ground-truth label k and $N_{\text{total}} = \sum_{j=1}^{17} N_j$ is the total number of training segments in AF17. During training, early stopping was applied, and the optimal model was selected based on the maximum Macro-F1 score achieved on the internal validation set.

At the window level, we report the F1 score and the area under the precision–recall curve (AUPRC). F1 is the harmonic mean of precision and recall and summarizes the balance between missed detections and false alarms. AUPRC describes how well the model trades off recall and precision under class imbalance.

At the event level, we adopt a unified post-processing configuration with $K = 5$ for the K-consecutive rule, onset threshold $\tau_{\text{on}} = \tau^*$, a merging gap of 2 s, and a refractory period of 5 s after each detected event. The metrics include Event Sen, positive predictive value (PPV), FA/h, and Delay. Sensitivity is the fraction of true falls that are detected. PPV is the fraction of predicted fall events that correspond to true falls. FA/h is the number of false alarms normalized by monitoring time in hours and reflects the disturbance to users and caregivers. Latency is the time from the actual fall onset to the first alarm raised by the system.

Model complexity is measured by the number of trainable parameters, memory usage, and floating-point operations (FLOPs), which together characterize the storage requirement, deployment footprint, and computational cost.

3 Results

It is important to note that all window-level metrics reported in Sections 3.1 to 3.3 were evaluated exclusively on the original, unmodified isolated test segments from the UniMiB-SHAR dataset. The semi-synthetic continuous streams were uniquely constructed and applied only for the event-level evaluation in Section 3.4.

3.1 Performance Comparison Under Different Partition Protocols

To systematically evaluate the proposed model under different cross-subject settings, we consider two protocols: a fixed subject-wise 70:15:15 split for training, validation and test, and a leave-one-subject-out (LOSO) protocol. For both protocols, window-level metrics are computed on the AF17 and AF2.

Table 2 reports, under the fixed subject-wise split, the mean and standard deviation of Accuracy, Macro-F1 and AUPRC. Specifically, these metrics were computed across 5 independent runs. During these runs, the 70:15:15 subject-wise data partition remained strictly fixed. Consequently, the reported variance stems solely from the stochasticity of the training process, including different random weight initializations and mini-batch shuffling, rather than from varying the data splits.

As shown in Table 2, under the fixed subject-wise split protocol LFNET achieves a favourable trade-off between performance and complexity on both the multi-class task AF17 and the binary task AF2. On the more challenging AF17, all models perform noticeably worse than on AF2. ResNet attains the highest F1 score of 67.5% and an AUPRC of 69.7%. LFNET, with only 0.18M parameters and 2.39M FLOPs, reaches an F1 of 69.1% and an AUPRC of 70.7%, essentially matching ResNet in F1 while improving AUPRC by about 1 percentage point. At the same time, LFNET uses roughly one twenty-first of the parameters and memory of ResNet, and about one eleventh of its computational cost, and also compares favourably

Table 2 Comparison of LFNET and existing models on AF2 and AF17 under the fixed subject-wise split

Model	Accuracy	Macro-F1	AUPRC	Params (M)	Memory (M)	FLOPs (M)
AF17						
CNN	81.2% ± 0.5%	65.2% ± 0.6%	66.2% ± 0.5%	0.52	2.02	6.88
Resnet	83.2% ± 0.4%	67.5% ± 0.5%	69.7% ± 0.4%	3.85	14.70	27.28
Lstm	58.9% ± 1.3%	42.7% ± 1.4%	50.3% ± 1.1%	0.14	0.54	21.03
Vit	79.5% ± 0.6%	61.5% ± 0.7%	63.2% ± 0.5%	0.56	2.16	9.95
Swin	80.1% ± 0.5%	63.2% ± 0.6%	60.6% ± 0.6%	0.32	1.25	11.38
LFNET	84.0% ± 0.3%	69.1% ± 0.5%	70.7% ± 0.4%	0.18	0.70	2.39
AF2						
CNN	97.6% ± 0.2%	96.9% ± 0.2%	96.1% ± 0.3%	0.52	1.98	6.88
Resnet	98.8% ± 0.1%	98.0% ± 0.1%	98.1% ± 0.2%	3.85	14.67	27.23
Lstm	94.2% ± 0.5%	93.0% ± 0.5%	96.3% ± 0.4%	0.14	0.53	21.03
Vit	97.7% ± 0.2%	96.9% ± 0.2%	97.9% ± 0.2%	0.56	2.16	9.95
Swin	97.5% ± 0.2%	96.7% ± 0.2%	98.0% ± 0.2%	0.32	1.24	11.38
LFNET	98.9% ± 0.1%	98.1% ± 0.1%	98.7% ± 0.1%	0.18	0.67	2.39

with other CNN, ViT and Swin variants, offering a markedly better performance–efficiency balance for resource-constrained scenarios.

On AF2, overall performance is high and several models achieve F1 and AUPRC values above 96%. ResNet again performs strongly (F1 98.0%, AUPRC 98.1%), whereas LFNET, with the same lightweight configuration (0.18M parameters, 2.39M FLOPs), further improves F1 to 98.1% and AUPRC to 98.7%, attaining the best results among all models. By contrast, LSTM also has a relatively small parameter count, but its F1 is only 42.7% on AF17 and 93.0% on AF2, with substantially higher FLOPs, indicating that purely recurrent temporal modelling struggles to balance complex postural patterns and computational efficiency. Overall, across both task settings, LFNET matches or surpasses larger convolutional and Transformer-based models while using a fraction of their parameters and compute, underscoring the effectiveness and practical value of the proposed architecture for smartphone-based fall detection.

To more comprehensively assess window-level fall-detection performance, we further compare six architectures CNN, LSTM, ResNet, Swin, ViT and LFNET under a common LOSO partition in both the AF17 and AF2. As illustrated in Fig. 4, the two configurations differ markedly in both task difficulty and discriminative power.

In the more challenging AF17 multi-class setting, all models show only moderate window-level performance, with F1 scores roughly between 0.46 and 0.66 and AUPRC values between 0.56 and 0.75. The large spread across subjects indicates that, under strong class imbalance and marked cross-subject variability, the models still struggle to reliably distinguish different fall types from high-dynamic daily activities.

LSTM performs worst, while CNN and Transformer-based models are in the middle, and ResNet and LFNET perform best. LFNET attains approximately 0.65 F1 and 0.75 AUPRC, with its subject-wise distribution clearly higher than the other baselines.

When the task is reduced to AF2, window-level performance is nearly saturated: mean F1 scores range from 0.94–0.98 and AUPRC values from 0.97–0.998, with much smaller performance gaps between models. LFNET remains slightly ahead, achieving 0.982 ± 0.026 F1 and 0.998 ± 0.006 AUPRC.

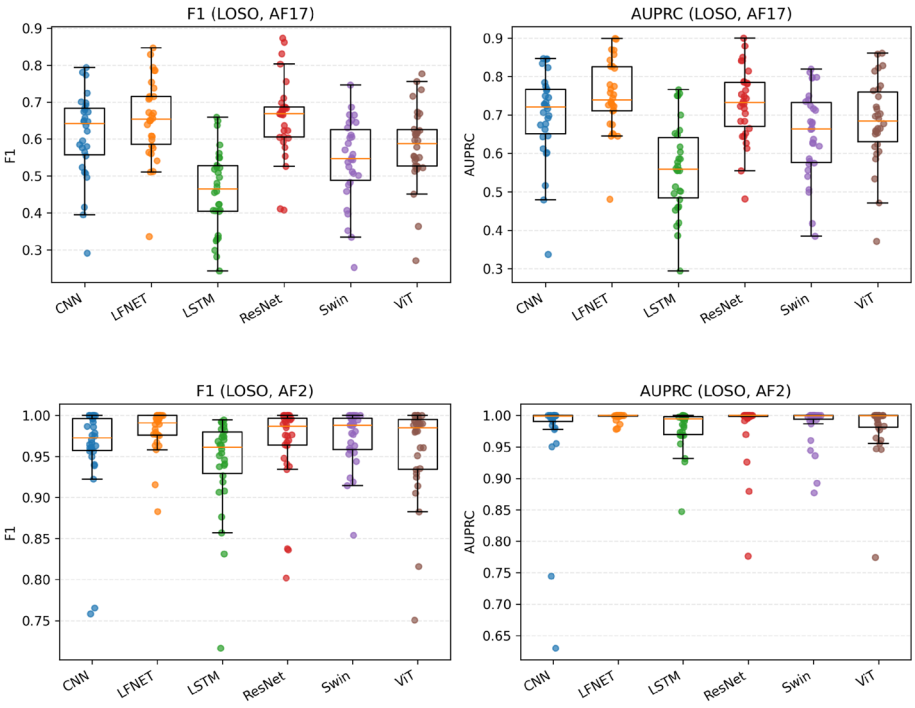


Fig. 4 The comparison results of the model under the LOSO classification

Overall, AF2 results suggest that coarse-grained fall versus non-fall detection is almost solved at the window level, whereas the lower AF17 performance and larger cross-subject variance underscore the need for multi-class formulations and finer event-level metrics such as FA/h and Delay in realistic evaluations.

3.2 Model Lightweighting and Multi-Channel Ablation

Having fixed the task settings and overall architecture, we next conduct a systematic ablation study on LFNET to quantify the individual contributions of its design components to both performance and complexity.

As summarized in Table 3, we first examine structural ablations under the fixed subject-wise split. Using CNN as a baseline on AF17, F1 and AUPRC reach 65.2% and 66.2%. Adding ECA channel attention (CNN+ECA) raises them to 68.2% and 69.3% with only a slight increase in parameters and memory, indicating that lightweight channel re-calibration substantially improves discrimination of complex activities. Replacing standard convolutions with depthwise separable convolutions (Dwconv) reduces parameters and FLOPs to 0.18 M and about 2.4 M (roughly one third of CNN), but F1 and AUPRC drop to 64.5% and 66.9%, showing that aggressive lightweighting alone incurs a modest accuracy loss. In contrast, LFNET, which combines multi-scale depthwise separable convolutions with ECA at essentially the same complexity as Dwconv, achieves 69.1% F1 and 70.7% AUPRC, outperforming both CNN and CNN+ECA and effectively recovering the performance sacrificed by pure light-weighting.

Table 3 Ablation Experiment Results under Fixed Partitioning

Model	F1	AUPRC	Params (M)	Memory (M)	FLOPs (M)
AF17					
CNN	65.2%	66.2%	0.52	2.02	6.88
CNN+ECA	68.2%	69.3%	0.57	2.18	6.95
Dwconv	64.5%	66.9%	0.18	0.70	2.38
LFNET	69.1%	70.7%	0.18	0.70	2.39
LFNET(xyz)	64.0%	68.2%	0.18	0.70	2.39
LFNET(mag)	62.8%	64.3%	0.18	0.70	2.39
AF2					
CNN	96.9%	96.1%	0.52	1.98	6.88
CNN+ECA	98.7%	98.8%	0.56	2.15	6.94
Dwconv	96.7%	98.5%	0.18	0.67	2.37
LFNET	98.1%	98.7%	0.18	0.67	2.39
LFNET(xyz)	97.6%	97.8%	0.18	0.67	2.39
LFNET(mag)	97.5%	97.6%	0.18	0.67	2.39

This table shows the ablation experiment results for different models under fixed subject-wise partitioning. The performance metrics include F1 score, AUPRC, parameters count (Params), memory usage (Memory), and FLOPs. LFNET variants with different input channel configurations are also compared

Input-channel ablations further highlight the benefit of the augmented multi-channel design. On AF17, removing the magnitude channel (LFNET xyz) yields 64.0% F1 and 68.2% AUPRC, while using only the magnitude (LFNET mag) degrades performance to 62.8% and 64.3%. Both are clearly inferior to the full multi-channel LFNET (xyz+mag), indicating that the axes and magnitude provide complementary information about posture and impact. On AF2, all models perform strongly, but the same trends persist: CNN improves from 96.9%/96.1% (F1/AUPRC) to 98.7%/98.8% with ECA, Dwconv maintains high scores despite substantial reductions in parameters and FLOPs, and LFNET matches Dwconv in complexity while reaching 98.1% F1 and 98.7% AUPRC, slightly ahead of reduced-channel variants. Overall, the combination of channel attention, depthwise separable convolutions, and multi-channel input yields a favourable balance between accuracy and efficiency.

To complement the quantitative metrics, Fig. 5 shows window-level confusion matrices for the CNN baseline and LFNET on AF17 and AF2 under the fixed subject-wise split.

On AF17, the CNN confusion matrix exhibits many prominent off-diagonal entries between multiple ADL classes and fall classes. Highly dynamic ADL segments are often misclassified as falls, and true falls are frequently confused with neighbouring ADL categories, which constrains overall recall and AUPRC. In contrast, LFNET produces a much more concentrated main diagonal: recall for fall-related classes is consistently higher and cross-confusion among ADL classes is clearly reduced, indicating that the combination of multi-scale depthwise separable convolutions and channel attention strengthens class-level separation between fall and non-fall patterns.

On AF2, both models achieve high sensitivity and specificity, but LFNET further reduces false negatives in the fall class, reflected by a larger "fall → fall" cell and a smaller "fall → non-fall" cell. This suggests that, under a risk structure where missed falls are more critical than false alarms, LFNET is better at correctly capturing true fall events without substantially

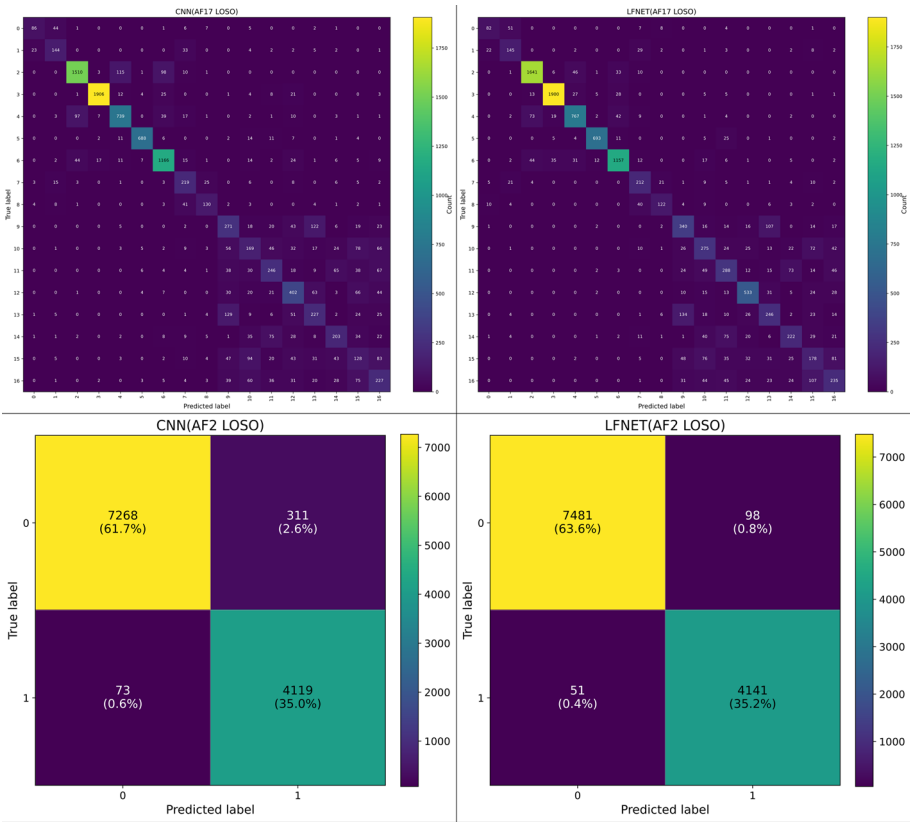


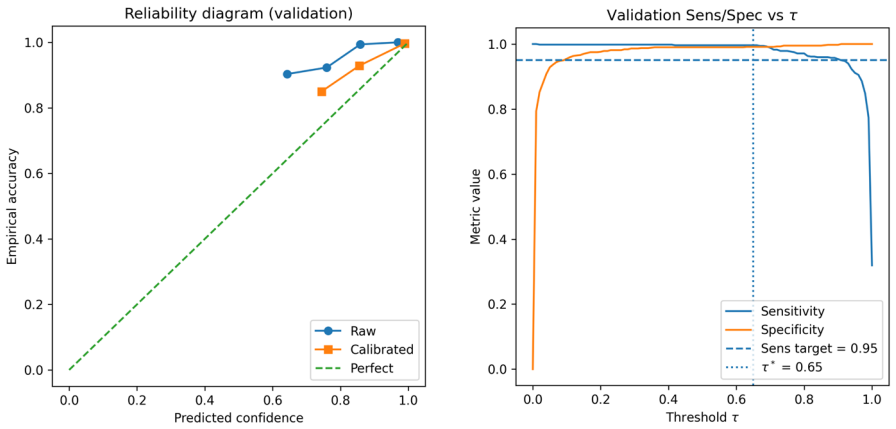
Fig. 5 Comparison chart of confusion matrices for LFNET and CNN models

increasing spurious alerts, providing confusion-matrix evidence for the improved robustness of the proposed architecture.

3.3 Probability Calibration and Threshold Analysis

To assess the effectiveness of temperature scaling and threshold search, we analyse probability calibration and decision thresholds on AF2, as shown in Fig. 6. Fig. 6a depicts the relationship between predicted confidence and empirical accuracy on the validation set before and after calibration. Most samples fall in the mid-to-high confidence range (approximately 0.5–1.0), and the uncalibrated model already achieves accuracies above 0.9 in most confidence bins, indicating strong separability on AF2. After temperature scaling, the reliability curve becomes smoother and adheres more closely to the diagonal in regions with sufficient samples, indicating that predicted probabilities more accurately reflect observed frequencies, without compromising overall classification performance.

Fig. 6b plots sensitivity and specificity as functions of the decision threshold τ under the fixed temperature T^*. For very low τ, sensitivity is near 1 while specificity is low; as τ increases slightly, specificity rises quickly, and both metrics remain close to 1 over a relatively wide threshold range, reflecting good separability between fall and non-fall samples



(a) Reliability curves before and after temperature scaling on AF2 validation data.

(b) Sensitivity and specificity versus decision threshold under calibrated probabilities.

Fig. 6 Calibration and threshold behaviour of LFNET on AF2

Table 4 Comparison of event-level performance between CNN and LFNET in simulated long-term monitoring scenarios.

Model	Event Sens (%)	FA/h	PPV (%)	Delay (s)
CNN	90.1	1.4	82.3	1.4
LFNET	96.4	0.2	89.1	0.8

in probability space. When τ approaches 1, specificity stays high but sensitivity drops sharply, implying a high risk of missed falls. Reflecting the risk preference of "avoiding missed falls over reducing false alarms," we impose a sensitivity lower bound of 0.95 and search over feasible thresholds that satisfy this constraint. Among them, we select τ^* at the point that maximizes AUPRC (vertical dashed line in Fig. 6b). At this operating point, sensitivity remains well above 0.95 and specificity is still close to 1, indicating that τ^* achieves effective control of false alarms while strictly limiting missed detections.

3.4 Event-Level Performance on Semi-Synthetic Streams

In the semi-synthetic stream experiments, we further compare the event-level performance of the CNN and LFNET in a simulated long-term monitoring scenario. Specifically, for each of five subjects we generate three streams of approximately 1 hour and insert multiple time-stamped fall events into each stream. We then compute Event Sens, FA/h, PPV, Delay. The results are summarized in Table 4.

Under the same threshold and post-processing configuration, LFNET clearly outperforms CNN at the event level. Specifically, CNN achieves an Event Sens of 90.1%, FA/h of 1.4, PPV of 82.3%, and a median detection delay of about 1.4 s. In contrast, LFNET increases Event Sens to 96.4%, while simultaneously reducing FA/h to about 0.2, raising PPV to 89.1%, and shortening the median delay to roughly 0.8 s. These results indicate that, in long-term monitoring conditions closer to real-world use, LFNET not only reduces missed detections and false alarms but also triggers alerts more quickly, further confirming its overall advantage over the CNN baseline at the event level.

4 Discussion

4.1 Summary of Main Findings

This work targets smartphone-based fall detection using only a single waist-worn accelerometer and jointly examines lightweight architecture design, multi-channel input, and probability calibration with event-level post-processing. On UniMiB-SHAR, LFNET achieves strong performance on both AF2 and AF17 while using far fewer parameters and FLOPs than ResNet and Transformer-based baselines, yet attaining comparable or superior window-level F1 and AUPRC. The gains are particularly evident on the more challenging AF17 task, suggesting that LFNET learns finer-grained temporal–postural patterns that benefit not only fall versus non-fall discrimination but also more detailed activity and risk-state characterization.

Temperature scaling and threshold search further improve probability calibration without degrading classification accuracy. Event-level evaluation on semi-synthetic continuous streams shows that, under a common operating point, LFNET achieves higher event sensitivity, lower FA/h, and shorter detection delay than the CNN baseline, indicating better overall suitability for long-term deployment.

4.2 Impact of Multi-Channel and Lightweight Design

Multi-channel ablations show that using only tri-axial components or only the magnitude leads to consistent degradation on AF17 and AF2, whereas combining xyz and magnitude yields the best results. Axes information captures posture and orientation changes, whereas magnitude encodes impact intensity. Their fusion improves discrimination between highly dynamic but non-fall ADL and true falls with strong impacts, and notably reduces confusion among hard-to-distinguish activities on AF17. Although the target deployment task is binary fall detection, the gains on AF17 indicate that LFNET has a general capacity for modelling complex activity patterns, which helps suppress false alarms from "high-risk ADL" and supports potential extensions to daily activity monitoring and rehabilitation assessment.

On the architectural side, depthwise separable convolutions substantially reduce parameters and FLOPs, and ECA channel attention yields stable performance gains at negligible extra cost. LFNET combines multi-scale depthwise separable convolutions with channel attention and, at a complexity close to a pure depthwise design, surpasses CNN and its attention variant on AF17 and AF2 and approaches or exceeds larger ResNet and Transformer models. This demonstrates that carefully designed lightweight architectures can deliver near "heavy-model" accuracy on resource-constrained platforms such as smartphones while retaining fine-grained recognition capability.

4.3 Calibration, Thresholds, and Event-Level Behaviour

Window-level F1 alone does not reflect how well predicted probabilities support practical alarm strategies. In this study, temperature scaling improves the agreement between confidence scores and empirical frequencies, and a high-sensitivity constraint on the validation set guides selection of the decision threshold. Combined with fixed event-level parameters, this procedure turns calibrated window-level outputs into operational alarms.

On semi-synthetic streams, LFNET improves event sensitivity, FA/h, and detection delay simultaneously compared with CNN at the same operating point. This indicates that the combination of calibration, high-sensitivity thresholding, and event-level post-processing

helps translate window-level metrics into more engineering-relevant alarm performance for smartphone-based, long-term monitoring.

4.4 Dataset Differences and Limitations

UniMiB-SHAR consists of scripted laboratory actions stored as isolated 3 s segments and therefore differs from the distribution of continuous, in-the-wild smartphone data. The semi-synthetic streams constructed in this work approximate continuous monitoring but still deviate from real-world conditions in fall frequency and activity diversity. In practice, collecting continuous, long-term fall data in unconstrained environments—especially from high-risk older adults—is exceptionally difficult and involves significant ethical and physical safety risks. Consequently, our semi-synthetic stream approach serves as a necessary methodological compromise. As such, the reported event-level FA/h and latency should be strictly regarded as optimistic estimates under controlled settings, which demonstrate the model's potential for long-term applications rather than its definitive real-world performance.

Moreover, the study is limited to a single dataset, wearing location, and sensor modality. Future work should include collecting continuous data in homes and care facilities, re-tuning post-processing parameters for these environments, and investigating multi-sensor fusion, self-supervised pretraining, and cross-device or cross-scenario transfer to further strengthen robustness and generalizability in real-world fall-detection applications.

5 Conclusion

This study proposed and systematically evaluated a lightweight deep learning framework for fall detection using a single waist-worn smartphone accelerometer. The method combines a multi-channel input with a lightweight LFNET backbone based on multi-scale depthwise separable convolutions and channel attention. On UniMiB-SHAR, LFNET was evaluated on both AF2 and AF17 under a fixed subject-wise split and leave-one-subject-out (LOSO), and achieved window-level F1 and AUPRC comparable to or better than larger convolutional and Transformer models, while requiring only a small fraction of their parameters and FLOPs. This indicates that the proposed architecture provides strong discrimination and good generalization on resource-constrained devices.

We further applied temperature scaling and threshold search to calibrate output probabilities and select an operating point aligned with the high-sensitivity requirements of fall detection. Combined with event-level post-processing on semi-synthetic continuous streams, this yielded favourable event-level sensitivity, FA/h and detection delay. Compared with a CNN baseline, LFNET achieved higher event detection rates while substantially reducing false alarms and latency, demonstrating practical advantages in semi-synthetic continuous monitoring evaluations.

Future work will collect long-duration continuous data in real home and care settings to recalibrate temperature, thresholds and event-level parameters, and to verify robustness and transferability across populations, device form factors and usage scenarios. Extending the framework with multi-sensor fusion, self-supervised pretraining and uncertainty estimation is a promising direction toward broader applications in activity monitoring and health risk early warning.

Author Contributions Conceptualization, Z.Y. and Q.M.; methodology, Z.Y., C.B. and K.X.; software, Z.Y. and D.L.; validation, Z.Y., Q.M., Y.X., L.L. and L.T.; formal analysis, Z.Y., Q.M. and L.L.; investigation, D.L.,

Y.X. and L.L.; resources, Z.R. and K.X.; data curation, Z.Y. and Q.M.; writing—original draft preparation, Z.Y.; writing—review and editing, Q.M., Z.Y., Z.R. and L.L.; visualization, Z.Y. and D.L.; supervision, C.B. and Z.R.; project administration, Z.R. and C.B.; funding acquisition, Z.R. All authors have read and agreed to the published version of the manuscript.

Funding This research was funded by the National Key R&D Program of China (Grant Nos.2024YFC2707903, 2023YFC3603502)
and the National Natural Science Foundation of China (Grant Nos. 11372223, 11102135).

Data Availability The data that support the findings of this study can be requested from the corresponding author.

Declarations

Competing interests The authors declare that they have no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Chen M, Wang H, Yu L, Yeung EHK, Luo J, Tsui K-L, Zhao Y (2022) A systematic review of wearable sensor-based technologies for fall risk assessment in older adults. Sensors 22(18):6752
2. Chen Y, Dai F, Huang S, Qi D, Peng C, Zhang A, Wang Y, Gu Y, Guo J (2025) Global, regional, and national burden of falls among older adults: findings from the global burden of disease study 2021 and projections to 2040. npj Aging 11(1):85
3. Eost-Telling C, Yang Y, Norman G, Hall A, Hanratty B, Knapp M, Robinson L, Todd C (2024) Digital technologies to prevent falls in people living with dementia or mild cognitive impairment: a rapid systematic overview of systematic reviews. Age Ageing 53(1):238
4. Gorce P, Jacquier-Bret J (2025) Fall detection in elderly people: a systematic review of ambient assisted living and smart home-related technology performance. Sensors 25(21):6540
5. Harwood RH (2022) The world falls guideline, vol 51. Oxford University Press, Oxford, UK, p 229
6. Heitmayer M, Lahlou S (2021) Why are smartphones disruptive? an empirical study of smartphone use in real-life contexts. Comput Hum Behav 116:106637
7. Khan ZY, Niu Z (2021) Cnn with depthwise separable convolutions and combined kernels for rating prediction. Expert Syst Appl 170:114528
8. Khatun MA, Yousuf MA, Ahmed S, Uddin MZ, Alyami SA, Al-Ashhab S, Akhdar HF, Khan A, Azad A, Moni MA (2022) Deep cnn-lstm with self-attention model for human activity recognition using wearable sensor. IEEE J Transl Eng Health Med 10:1–16
9. Kristoffersson A, Lindén M (2022) A systematic review of wearable sensors for monitoring physical activity. Sensors 22(2):573
10. Li X, Zhao P, Wu M, Chen Z, Zhang L (2021) Deep learning for human activity recognition. In: handbook of artificial intelligence in healthcare vol. 444, pp. 214–216. Elsevier, Amsterdam, The Netherlands
11. Micucci D, Mobilio M, Napoletano P (2017) Unimib-shar: a dataset for human activity recognition using acceleration data from smartphones. Appl Sci 7(10):1101
12. Noble WS (2006) What is a support vector machine? Nat Biotechnol 24(12):1565–1567
13. Ponce H, Martinez-Villasenor L, Nunez-Martinez J (2020) Sensor location analysis and minimal deployment for fall detection system. IEEE Access 8:166678–166691

14. Site A, Lohan ES, Jolanki O, Valkama O, Hernandez RR, Latikka R, Alekseeva D, Vasudevan S, Afolaranmi S, Ometov A (2022) Others: managing perceived loneliness and social-isolation levels for older adults: a survey with focus on wearables-based solutions. Sensors 22(3):1108
15. Sousa Lima W, Souto E, El-Khatib K, Jalali R, Gama J (2019) Human activity recognition using inertial sensors in a smartphone: an overview. Sensors 19(14):3213
16. Straczkiewicz M, James P, Onnela J-P (2021) A systematic review of smartphone-based human activity recognition methods for health research. NPJ Digital Medicine 4(1):148
17. Subramaniam S, Faisal AI, Deen MJ (2022) Wearable sensor systems for fall risk assessment: a review. Frontiers in Digital Health 4:921506
18. Suh S, Rey VF, Lukowicz P (2023) Tasked: transformer-based adversarial learning for human activity recognition using wearable sensors via self-knowledge distillation. Knowl-Based Syst 260:110143
19. Toyabe S-i (2010) World health organization fracture risk assessment tool in the assessment of fractures after falls in hospital. BMC Health Serv Res 10(1):106
20. Warrington DJ, Shortis EJ, Whittaker PJ (2021) Are wearable devices effective for preventing and detecting falls: an umbrella review (a review of systematic reviews). BMC Public Health 21(1):2091
21. Yang Y, Wang H, Jiang R, Guo X, Cheng J, Chen Y (2022) A review of iot-enabled mobile healthcare: technologies, challenges, and future trends. IEEE Internet Things J 9(12):9478–9502
22. Young F, Mason R, Morris RE, Stuart S, Godfrey A (2023) Iot-enabled gait assessment: the next step for habitual monitoring. Sensors 23(8):4100
23. Zhang Y, Zhan Q, Ma Z (2024) Efficientnet-eca: a lightweight network based on efficient channel attention for class-imbalanced welding defects classification. Adv Eng Inform 62:102737

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Zhiyuan Yang[1,4,5] · Liang Li[5] · Zhiqiang Ren[4,5] · Qinghua Meng[1] · Yijia Xue[1,4,5] · Can Bu[3,4,5] · Liangze Tao[3,4,5] · Kai Xiong[4,5] · Dongren Liu[1,2]

✉ Liang Li
liliang@ciss.cn

✉ Qinghua Meng
mqh13102231178@tjus.edu.cn

✉ Dongren Liu
2020200006@stu.tjus.edu.cn

Zhiyuan Yang
yzyyyds@bsu.edu.cn

Zhiqiang Ren
renzhiqiang@greatteam.org.cn

Yijia Xue
yijia.xue@tjus.edu.cn

Can Bu
bucan@seu.edu.cn

Liangze Tao
tao_liangze@seu.edu.cn

Kai Xiong
kaixiong@zut.edu.cn

1 Tianjin University of Sport, Tuanbo New Town, Jinghai District, 301617 Tianjin, China

2 Shandong Sports Rehabilitation Research Center, Jinan 250102, China

[3] School of Automation, Southeast University, Nanjing 210096, China

[4] Greatteam Smart Sport Technology Innovation Center(Beijing) Co, Ltd., Greatteam Smart Sport Technology Innovation Center, Beijing 100061, China

[5] China Institute of Sport Science, Beijing 100061, China